

Vignesh Narayanan

Assistant Professor, Department of Computer Science and Engineering
University of South Carolina

Core Faculty, AI Institute (AIISC) | Core Faculty, Carolina Autism & Neurodevelopment (CAN) Research Center

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Research Interests: Dynamical systems and networks, Control theory, Artificial intelligence for collaborative autonomy, and Computational neuroscience

Professional Experience

Assistant Professor, Department of Computer Science and Engineering *Aug 2021 – Present*

University of South Carolina, Columbia, South Carolina, USA

- Core faculty, AI Institute (AIISC) *Aug 2021 – Present*
- Core faculty, Carolina Autism and Neurodevelopment (CAN) Research Center *Aug 2021 – Present*

Postdoctoral Research Associate, Department of Electrical and Systems Engineering *Sep 2017 – Jul 2021*

Washington University in St. Louis, St. Louis, Missouri, USA

- Affiliate, Applied Mathematics Lab *Sep 2017 – Jul 2021*
- Affiliate, Brain Dynamics and Control Research Group *Sep 2017 – Jul 2019*

Graduate Research Assistant, Department of Electrical and Computer Engineering *Aug 2014 – Jul 2017*

Missouri University of Science and Technology, Rolla, Missouri, USA

Education

Ph.D. in Electrical Engineering *Jul 2017*

Missouri University of Science and Technology, Rolla, Missouri, USA

M.Tech. in Control Systems *Jun 2014*

National Institute of Technology, Kurukshetra, Haryana, India

B.Tech. in Electrical and Electronics Engineering *Jun 2012*

SASTRA University, Thanjavur, Tamil Nadu, India

Coauthored Textbook

1. Jagannathan Sarangapani, Vignesh Narayanan, and Avimanyu Sahoo, “Optimal Event-Triggered Control Using Adaptive Dynamic Programming,” CRC Press, 2024.

Published Journal Articles

1. Erik Connerty, Ethan Evans, Gerasimos Angelatos, and Vignesh Narayanan, “A NISQ hardware demonstration of chaotic system prediction with quantum echo-state networks,” *Communications Physics*, 2026, in press.
2. Julia Yurkovic-Harding, Vignesh Narayanan, and Jessica Bradshaw, “Social behavior forecasts moment-to-moment changes in RSA in infants with autism,” *Developmental Science*, 2026, in press.
3. Ritirupa Dey, Avimanyu Sahoo, and Vignesh Narayanan, “State-constrained online adaptive control for robotic manipulators,” *IEEE Control Systems Letters*, 2025.
4. Ritirupa Dey, Avimanyu Sahoo, and Vignesh Narayanan, “Safe data-enabled control of human-in-the-loop robotic manipulator systems,” *IEEE Transactions on Industrial Informatics*, 2025, in press.
5. Bharath Muppasani, Nitin Gupta, Vishal Pallagani, Biplav Srivastava, Raghava Mutharaju, Michael N. Huhns, and Vignesh Narayanan, “Building a planning ontology to represent and exploit planning knowledge and its applications,” *Discover Data*, 2025, vol. 3(1): 55.
6. Vignesh Narayanan, Lawrence Bordoh, Istvan Z. Kiss, and Jr-Shin Li, “Inferring networks of chemical reactions by curvature analysis of kinetic trajectories,” *Physical Chemistry Chemical Physics*, The Royal Society of Chemistry, 2025 (Featured on the cover page).
7. Yuan-Hung Kuan, Vignesh Narayanan, and Jr-Shin Li, “An iterative reservoir computing network approach to irregular time-series data reconstruction,” *IEEE Transactions on Neural Networks and Learning Systems*, 2025, in press.
8. Vignesh Narayanan, Wei Zhang, and Jr-Shin Li, “Duality of ensemble systems through moment representations,” *IEEE Transactions on Automatic Control*, vol. 69(10), pp. 7270–7276, Oct. 2024.
9. Chathurangi Shyalika, Kaushik Roy, Renjith Prasad, Fadi El Kalach, Yuxin Zi, Priya Mittal, Vignesh Narayanan, Ramy Harik, and Amit Sheth, “RI2AP: Robust and interpretable 2D anomaly prediction in assembly pipelines,” *Sensors*, vol. 24(10), pp. 3244, 2024.
10. Maxwell Geiger, Vignesh Narayanan, Sarangapani Jagannathan, “Optimal trajectory tracking for uncertain linear discrete-time systems using time-varying Q-learning,” *International Journal of Adaptive Control and Signal Processing*, vol. 38(7), pp. 2340–2368, 2024.
11. Yao-Chi Yu, Vignesh Narayanan, and Jr-Shin Li, “Moment-based reinforcement learning for ensemble control,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35(9), pp. 12653–12664, Sept. 2024.
12. Krishnan Raghavan, Vignesh Narayanan, and Sarangapani Jagannathan, “Cooperative deep Q-learning framework for environments providing image feedback,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 35(7), pp. 9267–9276, July 2024.
13. Rohollah Moghadam, Vignesh Narayanan, and Sarangapani Jagannathan, “Event-triggered optimal adaptive control of partially unknown linear continuous-time systems with state delay,” *IEEE Systems, Man, and Cybernetics: Systems*, vol. 53(6), pp. 3324–3337, June 2023.
14. Wei Miao, Vignesh Narayanan, and Jr-Shin Li, “Interpretable design of reservoir computing networks using realization theory,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34(9), pp. 6379–6389, Sept. 2023.
15. Vignesh Narayanan, Hamidreza Modares, Sarangapani Jagannathan, and Frank L. Lewis, “Event-driven off-policy reinforcement learning for control of interconnected systems,” *IEEE Transactions on Cybernetics*, vol. 52(3), pp. 1936–1946, March 2022.
16. Liang Wang, Vignesh Narayanan, Yao-Chi Yu, Yikyung Park, and Jr-Shin Li, “A two-stage hierarchical clustering method for structured temporal sequence data,” *Knowledge and Information*

Systems, vol. 63, pp. 1627–1662, 2021.

17. Vignesh Narayanan, Hamidreza Modares, and Sarangapani Jagannathan, “Optimal event-triggered control of input-affine nonlinear interconnected systems using multi-player games,” *International Journal of Robust and Nonlinear Control*, vol. 31, pp. 950–970, 2021.
18. Wei-Cheng Jiang, Vignesh Narayanan, and Jr-Shin Li, “Model learning and knowledge sharing for cooperative multiagent systems in stochastic environment,” *IEEE Transactions on Cybernetics*, vol. 51(12), pp. 5717–5727, Dec. 2021.
19. Wei Miao, Vignesh Narayanan, and Jr-Shin Li, “Parallel residual projection: a new paradigm for solving linear inverse problems,” *Nature Scientific Reports*, vol. 10(1), pp. 1–10, 2020.
20. Avimanyu Sahoo and Vignesh Narayanan, “Differential-game for resource aware approximate optimal control of large-scale nonlinear systems with multiple players,” *Neural Networks*, Elsevier, vol. 124, pp. 95–108, 2020.
21. Vignesh Narayanan, Jr-Shin Li, and Shinung Ching, “Biophysically interpretable inference of single neuron dynamics,” *Journal of Computational Neuroscience*, Springer, vol. 47(1), pp. 61–76, 2019.
22. Avimanyu Sahoo and Vignesh Narayanan, “Optimization of sampling intervals for tracking control of nonlinear systems: A game theoretic approach,” *Neural Networks*, Elsevier, vol. 114, pp. 78–90, 2019.
23. Vignesh Narayanan, Avimanyu Sahoo, Sarangapani Jagannathan, and Koshy George, “Approximate optimal distributed control of nonlinear interconnected systems using event-triggered nonzero-sum games,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 30(5), pp. 1512–1522, 2018.
24. Avimanyu Sahoo, Vignesh Narayanan, and Sarangapani Jagannathan, “A min–max approach to event- and self-triggered sampling and regulation of linear systems,” *IEEE Transactions on Industrial Electronics*, vol. 66(7), pp. 5433–5440, 2018.
25. Vignesh Narayanan, Sarangapani Jagannathan, and Ram Kumar, “Event-sampled output feedback control of robotic manipulators using neural networks,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 30(6), pp. 1651–1658, 2017.
26. Vignesh Narayanan and Sarangapani Jagannathan, “Event-triggered distributed control of nonlinear interconnected systems using online reinforcement learning with exploration,” *IEEE Transactions on Cybernetics*, vol. 48(9), pp. 2510–2519, 2017.
27. Vignesh Narayanan and Sarangapani Jagannathan, “Event-triggered distributed approximate optimal state and output control of affine nonlinear interconnected systems,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 29(7), pp. 2846–2856, 2017.
28. Nathan Szanto, Vignesh Narayanan, and Sarangapani Jagannathan, “Event-sampled direct adaptive neural network output- and state-feedback control of uncertain strict-feedback system,” *IEEE Transactions on Neural Networks and Learning Systems*, vol. 29(5), pp. 1850–1863, 2017.
29. Vignesh Narayanan and Sarangapani Jagannathan, “Distributed adaptive optimal regulation of uncertain large-scale interconnected systems using hybrid Q-learning approach,” *IET Control Theory and Applications*, vol. 10(12), pp. 1448–1457, 2016.

Articles Under Review

30. Srikanth K S, Ramesh T K, Protik Nag, and Vignesh Narayanan, “Targeted layer recalibration in convolution neural networks: Enhancing concept alignment in image classification tasks,” *Pattern Recognition*.
31. Riturupa Dey, Bharath Chandra Muppasani, Avimanyu Sahoo, Biplav Srivastava, Vignesh Narayanan, “RePoC: Bridging the gap between Resource-efficient Planning and safe Optimal Control of non-holonomic wheeled mobile robots.”

Articles in Preparation

32. Protik Nag, Krishnan Raghavan, and Vignesh Narayanan, “Towards hierarchical continual learning to mitigate task ordering problem”.
33. Ritirupa Dey, Mahtab Noor Shaan, Bharath Muppasani, Avimanyu Sahoo, and Vignesh Narayanan, “Safe resource-efficient optimal control of nonholonomic wheeled mobile robots.”
34. Md Abir Hossen, Mohammad Ali Javidian, Vignesh Narayanan, Jason O’Kane, Pooyan Jamshidi, “Multi-objective multi-fidelity Bayesian optimization with causal priors”.

Published in Conference Proceedings

1. Bharath Muppasani, Risha Patel, Biplav Srivastava, Vignesh Narayanan, “HI-MAPF – Towards resource-efficient multi-agent deployment,” IROS, 2026.
2. Ritirupa Dey, Avimanyu Sahoo, and Vignesh Narayanan, “Output-constrained adaptive tracking control for safe robot manipulation,” WCCI, 2026.
3. Mahtab Noor Shaan, Vignesh Narayanan, and Avimanyu Sahoo, “Resource constrained safe reinforcement learning for lane following of wheeled mobile robots,” WCCI, 2026.
4. Bharath Muppasani, Ritirupa Dey, Biplav Srivastava, and Vignesh Narayanan, “maPO: An ontology for multi-agent path finding and its usage for explaining planner behavior,” AAAI-MAKE, 2026.
5. Avimanyu Sahoo and Vignesh Narayanan, “Integrated design of control and communication via discrete-time game formulation,” European Control Conference (ECC), 2026.
6. Saima Alam, Vignesh Narayanan, and Avimanyu Sahoo, “Thermal coupling identification in Lithium-ion battery packs,” European Control Conference (ECC), 2026.
7. Ritirupa Dey, Avimanyu Sahoo, and Vignesh Narayanan, “State-constrained online adaptive control for robotic manipulators,” American Control Conference, 2026 (accepted for presentation).
8. Bharath Muppasani, Ritirupa Dey, Biplav Srivastava, and Vignesh Narayanan, “OMEGA: An ontology-driven tool for explaining multi-agent path finding,” AAAI, 2026 (Demonstration track).
9. Arian Yousefian, Avimanyu Sahoo, and Vignesh Narayanan, “Online pseudospectral approach for system identification and state estimation,” IEEE Conference on Decision and Control (CDC), 2025.
10. Arian Yousefian, Avimanyu Sahoo, and Vignesh Narayanan, “A novel online system identification using Chebyshev polynomial basis of the second kind,” IEEE American Control Conference, 2025.
11. Bharath Muppasani, Protik Nag, Vignesh Narayanan, Biplav Srivastava, Michael Huhns, “Towards effective planning strategies for dynamic opinion networks,” Neural Information Processing Systems, 2024.
12. Bharath Muppasani, Nitin Gupta, Vishal Pallagani, Biplav Srivastava, Raghava Mutharaju, Michael N. Huhns, and Vignesh Narayanan, “A planning ontology to represent and exploit planning knowledge for performance efficiency,” CODS-COMAD, 2024.
13. Arian Yousefian, Avimanyu Sahoo, and Vignesh Narayanan, “Aperiodic sensing and data-driven system identification of nonlinear systems using Chebyshev pseudospectral approach,” IEEE Conference on Decision and Control (CDC), 2024.
14. Yuxin Zi, Kaushik Roy, Vignesh Narayanan, and Amit Sheth, “Exploring alternative approaches to language modeling for learning from data and knowledge,” AAAI-MAKE, 2024.
15. Kaushik Roy, Alessandro Oltramari, Yuxin Zi, Chathurangi Shyalika, Vignesh Narayanan, and Amit Sheth, “Causal event graph-guided language-based spatiotemporal question answering,” AAAI-MAKE, 2024.

16. Maxwell Geiger, Vignesh Narayanan, Sarangapani Jagannathan, "Optimal tracking of uncertain linear discrete-time systems using time-varying lifelong Q-learning," IEEE American Control Conference (ACC), 2024.
17. Bharath Muppasani, Vignesh Narayanan, Michael Huhns, and Biplav Srivastava, "Expressive and flexible simulation of information spread strategies in social networks using planning," AAAI, 2024 (Best technical demonstration award).
18. Revathy Venkataramanan, Kaushik Roy, Kanak Raj, Renjith Prasad, Yuxin Zi, Vignesh Narayanan, and Amit Sheth, "Cook-Gen: Robust generative modeling of cooking actions from recipes," IEEE Conference on Systems, Man and Cybernetics, 2023.
19. Kaushik Roy, Yuxin Zi, Manas Gaur, Jinendra Malekar, Qi Zhang, Vignesh Narayanan, and Amit Sheth, "Process knowledge-infused learning for clinician-friendly explanations," AAAI Second Symposium on Human Partnership with Medical Artificial Intelligence, 2023.
20. Shrey Gupta, Anmol Agarwal, Manas Gaur, Koushik Roy, Vignesh Narayanan, Ponnuram Kumaraguru, and Amit Sheth, "Learning to automate follow-up question generation using process knowledge for depression triage on Reddit posts," CLPsych@NAACL, 2022.
21. Vignesh Narayanan, Brett W. Robertson, Andrea Hickerson, Biplav Srivastava, and Bryant W. Smith, "Securing social media for seniors from information attacks: Modeling, detecting, intervening, and communicating risks," 2021 Third IEEE International Conference on Trust, Privacy and Security in Intelligent Systems and Applications (TPS-ISA), pp. 297–302. IEEE.
22. Avimanyu Sahoo, Vignesh Narayanan, and Qiming Zhao, "Adaptive gain observers for distributed state estimation of linear systems," IEEE American Control Conference (ACC), New Orleans, Louisiana, USA, 2021.
23. Avimanyu Sahoo, Vignesh Narayanan, and Qiming Zhao, "Finite-time adaptive optimal output feedback control of linear systems with intermittent feedback," IEEE Symposium Series on Computational Intelligence (SSCI), Canberra, Australia, 2020.
24. Yao-Chi Yu, Vignesh Narayanan, Shinung Ching, and Jr-Shin Li, "Learning to control neurons using aggregated measurements," IEEE American Control Conference (ACC), Denver, USA, pp. 4028–4033, 2020.
25. Wei Zhang, Vignesh Narayanan, and Jr-Shin Li, "Robust population transfer for coupled spin ensembles," IEEE Conference on Decision and Control (CDC), Nice, France, pp. 419–424, 2019.
26. Vignesh Narayanan, Jason Ritt, Jr-Shin Li, and Shinung Ching, "A learning framework for controlling spiking neural networks," IEEE American Control Conference (ACC), Philadelphia, USA, pp. 211–216, 2019.
27. Vignesh Narayanan, Sarangapani Jagannathan, and Rohollah Moghadam, "Optimality in event-triggered adaptive control of uncertain linear dynamical systems," AIAA Scitech Forum, San Diego, CA, USA, pp. 2187, 2019.
28. Avimanyu Sahoo and Vignesh Narayanan, "Event-based near optimal sampling and tracking control of nonlinear systems," IEEE Conference on Decision and Control (CDC), Miami Beach, FL, USA, pp. 55–60, 2018.
29. Vignesh Narayanan, Avimanyu Sahoo, and Sarangapani Jagannathan, "Approximate optimal distributed control of nonlinear interconnected systems using nonzero-sum games," IEEE Conference on Decision and Control (CDC), Miami Beach, FL, USA, pp. 2872–2877, 2018.
30. Vignesh Narayanan, Avimanyu Sahoo, and Sarangapani Jagannathan, "Optimal adaptive distributed control of linear interconnected systems," IEEE Symposium Series on Computational Intelligence (SSCI), Bangalore, India, pp. 1441–1446, 2018.
31. Avimanyu Sahoo, Vignesh Narayanan, and Sarangapani Jagannathan, "Event-triggered control of N-player nonlinear systems using nonzero-sum games," IEEE Symposium Series on Computational Intelligence (SSCI), Bangalore, India, pp. 1447–1452, 2018.

32. Vignesh Narayanan, Avimanyu Sahoo, and Sarangapani Jagannathan, "Optimal event-triggered control of nonlinear systems: A min-max approach," IEEE American Control Conference (ACC), Milwaukee, WI, USA, pp. 3441–3446, 2018.
33. Avimanyu Sahoo, Vignesh Narayanan, and Sarangapani Jagannathan, "Optimal event-triggered control of uncertain linear networked control systems: A co-design approach," IEEE Symposium Series on Computational Intelligence (SSCI), Honolulu, HI, 2017.
34. Avimanyu Sahoo, Vignesh Narayanan, and Sarangapani Jagannathan, "Optimal sampling and regulation of uncertain interconnected linear continuous time systems," IEEE Symposium Series on Computational Intelligence (SSCI), Honolulu, HI, 2017.
35. Haci Mehmet Guzey, Vignesh Narayanan, Sarangapani Jagannathan, Travis Dierks, and Levent Acar, "Distributed consensus-based event-triggered approximate control of non-holonomic mobile robot formations," IEEE American Control Conference (ACC), Seattle, WA, USA, pp. 3194–3199, 2017.
36. Nathan Szanto, Vignesh Narayanan, and Sarangapani Jagannathan, "Event-sampled control of quadrotor unmanned aerial vehicle," IEEE American Control Conference (ACC), Seattle, WA, USA, pp. 2956–2961, 2017.
37. Vignesh Narayanan and Sarangapani Jagannathan, "Online reinforcement with exploration for distributed control," International Joint Conference on Neural Networks (IJCNN), Anchorage, AK, USA, pp. 4022–4027, 2017.
38. Vignesh Narayanan and Sarangapani Jagannathan, "Approximate optimal distributed control of uncertain nonlinear interconnected systems with event-sampled feedback," IEEE Conference on Decision and Control (CDC), Las Vegas, NV, pp. 5827–5832, 2016.
39. Nathan Szanto, Vignesh Narayanan, and Sarangapani Jagannathan, "Event-sampled direct adaptive NN state-feedback control of uncertain strict-feedback system," IEEE Conference on Decision and Control (CDC), Las Vegas, NV, pp. 3395–3400, 2016.
40. Vignesh Narayanan and Sarangapani Jagannathan, "Event-sampled adaptive neural network control of robot manipulators," International Joint Conference on Neural Networks (IJCNN), Vancouver, BC, pp. 4941–4946, 2016.
41. Vignesh Narayanan and Sarangapani Jagannathan, "Distributed event-sampled approximate optimal control of interconnected affine nonlinear continuous-time systems," IEEE American Control Conference (ACC), Boston, MA, pp. 3044–3049, 2016.
42. Vignesh Narayanan and Sarangapani Jagannathan, "Distributed adaptive optimal regulation of uncertain large-scale linear networked control systems using Q-learning," IEEE Symposium Series on Computational Intelligence, pp. 587–592, 2015.
43. Vignesh Narayanan, Akhilesh Swarup, and Pulak Halder, "Design of autopilots for tactical aerospace vehicles: a comparative study," IFAC Proceedings, vol. 47(1), pp. 271–278, 2014.
44. Siddhardha Kedarisetty, Vignesh Narayanan, and Pulak Halder, "Autopilot design for flexible tactical aerospace vehicle using parameter plane technique," IFAC Proceedings, vol. 47(1), pp. 211–218, 2014.

Contributed Book Chapters

1. Vignesh Narayanan, Haci Mehmet Guzey, and Sarangapani Jagannathan, "Event sampled adaptive control of robot manipulators and mobile robot formations," *Adaptive control for robotic manipulators*, Editors: Dan Zhang, Bin Wei, CRC Press/Taylor & Francis Group, pp. 124–158, 2016.
2. Rohollah Moghadam, Sarangapani Jagannathan, Vignesh Narayanan, and Krishnan Raghavan, "Optimal adaptive control of partially uncertain linear continuous-time systems with state delay," *Handbook of Reinforcement Learning and Control*, Editors: Kyriakos G. Vamvoudakis, Yan Wan,

Contributed Magazine Articles

1. Avimanyu Sahoo, Vignesh Narayanan, Jagannathan Sarangapani, “Resource aware learning-based optimal control of cyber-physical systems,” *TC-CPS Newsletter*, vol. 6(1), pp. 24–34, March 2021.
2. Bharath Muppasani, Vishal Pallagani, Kausik Lakkaraju, Shuge Lei, Biplav Srivastava, Brett Robertson, Andrea Hickerson, and Vignesh Narayanan, “On safe and usable chatbots for promoting voter participation,” *AI Magazine*, vol. 44, pp. 240–247, 2023.

Preprints

1. Erik Connerty, Ethan Evans, Gerasimos Angelatos, and Vignesh Narayanan, “Predicting chaotic systems with quantum echo-state networks” (arXiv:2412.07910).
2. Bharath Muppasani, Protik Nag, Biplav Srivastava, and Vignesh Narayanan, “On generalized planning for controlling opinion networks: Interpreting human-AI dialog states and beliefs,” 2025.
3. Bharath Muppasani, Ritirupa Dey, Biplav Srivastava, and Vignesh Narayanan, “Scalable multi-agent path finding using collision-aware dynamic alert mask and a hybrid execution strategy” (arXiv:2510.09469).
4. Vignesh Narayanan, Wei Miao, and Jr-Shin Li, “Disentangling drift- and control-vector fields for interpretable inference of control-affine systems” (arXiv:2004.10954).
5. Vignesh Narayanan, Wei Zhang, and Jr-Shin Li, “Moment-based ensemble control” (arXiv:2009.02646).
6. Wei Zhang, Vignesh Narayanan, and Jr-Shin Li, “Dynamic function learning through control of ensemble systems”.

Posters and Abstracts

1. Julia Yurkovic-Harding, Vignesh Narayanan, John Richards, and Jessica Bradshaw, “Social behaviors forecast physiological regulation for infants with ASD and TD infants,” INSAR, 2025.
2. Bharath Muppasani, Protik Nag, Biplav Srivastava, and Vignesh Narayanan, “On generalized planning for controlling opinion networks: Interpreting human-AI dialog states and beliefs,” Workshop on Generalization in Planning, AAAI, Philadelphia, 2025.
3. Erik Connerty, Rae Jones, Protik Nag, Bharath Muppasani, Sai Teja Paladi, Vignesh Narayanan, Biplav Srivastava, and Michael Huhns, “Information competition simulator: A high-performance approach to modeling opinion dynamics in large populations,” DiscoverUSC, 2024.
4. Diksha Srishyla, Erik Connerty, Vignesh Narayanan, and Christian O’Reilly, “Systematic comparison of brain connectivity metrics for EEG,” DiscoverUSC, 2024.
5. Chathurangi Shyalika, Kaushik Roy, Renjith Prasad, Yuxin Zi, Priya Mittal, Fadi El Kalach, Vignesh Narayanan, Ramy Harik, and Amit Sheth, “RI2AP: Robust and interpretable 2D anomaly prediction in assembly pipelines,” DiscoverUSC, 2024 (Best poster award).
6. Deepa Tilwani, Christian O’Reilly, Raxitkumar Goswami, Nicholas Riccardi, Xuan Yang, Valerie L. Shalin, Vignesh Narayanan, Svetlana Shinkareva, Amit P. Sheth, and Rutvik Harshad Desai, “Predicting language outcomes from MRI post-stroke: A machine learning approach,” The Organization for Human Brain Mapping (OHBM), 29th Annual Meeting, July 2023.
7. Vignesh Narayanan, Bharath Muppasani, Sai Teja Paladi, Biplav Srivastava, and Michael Huhns, “Modeling and steering multi-dimensional opinion networks with aggregated measurements,” SIAM Conference on Applications of Dynamical Systems (DS23), Portland, Oregon, May 2023.

8. Vignesh Narayanan and Avimanyu Sahoo, “Optimal sampling, communication, and distributed control protocols via pseudo-spectral methods,” SIAM Conference on Applications of Dynamical Systems (DS23), Portland, Oregon, May 2023.
9. Avimanyu Sahoo, Geetika Vennam, and Vignesh Narayanan, “Neural network-based state estimation of lithium-ion batteries under internal faults,” SIAM Conference on Applications of Dynamical Systems (DS23), Portland, Oregon, May 2023.
10. “Event-driven approximate dynamic programming for feedback control,” Seminar in Advances in Computing Series, University of South Carolina, April 2022.
11. Jr-Shin Li, Wei Miao, and Vignesh Narayanan, “A scalable computational framework for solving large-scale linear regression problems,” The Second Joint SIAM/CAIMS Annual Meeting (AN20), Toronto, Canada, July 2020 (Accepted contributed lecture).
12. Vignesh Narayanan, Jr-Shin Li, and Shinung Ching, “Biophysically interpretable, model-free identification of neuronal dynamics,” Computational and Systems Neuroscience (Cosyne), Denver, CO, 2018.

Courses Taught

Designed and developed courses on *Neural Networks and their Applications* and *Optimization*.

- Fall 2021 – CSCE 790: Neural Networks and Their Applications
- Spring 2022 – CSCE 590: Optimization
- Fall 2022 – CSCE 790: Neural Networks and Their Applications
- Spring 2023 – CSCE 350H: Data Structures and Algorithms
- Fall 2023 – CSCE 790: Neural Networks and Their Applications
- Spring 2024 – CSCE 590: Optimization
- Fall 2024 – CSCE 790: Neural Networks and Their Applications
- Spring 2025 – CSCE 350H: Data Structures and Algorithms
- Fall 2025 – CSCE 584: Neural Networks and Their Applications (adopted as a permanent course)
- Spring 2026 – CSCE 590: Optimization

Scholarships and Awards

- Senior Member, IEEE, 2023
- Graduate Research Assistantship Award: August 2014 – 2017
- Graduate Research Assistantship Award (GATE): June 2012 – May 2014
- Gold Medal for top academic standing – M.Tech. Electrical Engineering in Control Systems

Lectures (L), Presentations (W), and Invited Talks (I)

Invited Talks (I)

- Workshop: “Pushing the Frontiers of Systems Theory in Quantum Science: Control, Learning, and Applications,” IEEE Conference on Decision and Control, December 2025.
- “Collaborative autonomy – Control of cyber-physical systems,” Seminar in Advances in Computing at the University of South Carolina, Columbia, August 2025.
- “Network inference for control and design of self-organization,” 11th International Conference on Engineering of Chemical Complexity (ECC11), Tokyo, Japan, July 2025.
- “Event-triggered control,” Workshop on Machine Learning and Deep Learning: Basics to Research Perspectives, Madras Institute of Technology Campus, Anna University, India, 2024.
- “Collaborative autonomy – From robotics, computational social science, to AI in mental health-

care,” Seminar in Advances in Computing at the University of South Carolina, Columbia, August 2024.

- “Systems theory of ensembles: Fundamentals, learning, and applications,” Workshop at the 62nd IEEE Conference on Decision and Control, 2023.
- “Event-driven approximate dynamic programming for feedback control,” Seminar in Advances in Computing at the University of South Carolina, Columbia, April 2022.
- “Dynamic learning and control for complex population systems and networks,” AIISC and USCAND Invited Seminar, University of South Carolina, March 2021.
- “Learning and control in dynamical systems and networks,” Dept. of EE, Invited Seminar, Indian Institute of Science, Bangalore, India, February 2021.
- “Event-triggered optimal adaptive control,” ESE Invited Talk, Washington University in St. Louis, May 2017.
- Vignesh Narayanan, Avimanyu Sahoo, and Sarangapani Jagannathan, “Optimal regulation of uncertain linear discrete-time systems using event-sampled Q-learning and adaptive dynamic programming,” 17th Yale Workshop on Adaptive and Learning Systems, 2015.

Workshop Presentations (W)

- Erik Connerty, Ethan Evans, Gerasimos Angelatos, and Vignesh Narayanan, “Predicting chaotic systems with quantum echo-state networks,” First International Workshop on Quantum Computing and Artificial Intelligence, AAAI, Philadelphia, 2025.
- “Dynamic function learning through control of ensemble systems,” The AAAI 2023 Workshop on When Machine Learning Meets Dynamical Systems, AAAI, Washington DC, Feb. 2023.
- “On safe and usable chatbots for promoting voter participation,” The AAAI 2023 Second Workshop on AI for Credible Elections: A Call To Action with Trusted AI, AAAI, Washington DC, Feb. 2023.
- Yuxin Zi, Kaushik Roy, Vignesh Narayanan, Manas Gaur, and Amit Sheth, “IERL: Interpretable ensemble representation learning – Combining crowd-sourced knowledge and distributed semantic representations,” KDD Workshop on Knowledge-infused Learning, 2023.
- Kaushik Roy, Yuxin Zi, Vignesh Narayanan, Manas Gaur, and Amit Sheth, “Knowledge-infused self-attention transformers,” KDD Workshop on Knowledge Augmented Methods for NLP, 2023.

Lectures (L)

- “Introduction to nonlinear dynamic systems,” Lecture for ESE 553: Nonlinear Dynamic Systems, Washington University in St. Louis.
- “Introduction to reinforcement learning for control,” Guest lecture for ESE 553: Nonlinear Dynamic Systems, Washington University in St. Louis.
- “Introduction to neural networks,” Lecture for EE 5320: Neural Networks Control and Application, Missouri University of Science and Technology (MST), Rolla.
- “Applications of neural networks in control,” Lecture for EE 5320: Neural Networks Control and Application, MST, Rolla.
- “Linear quadratic regulators and tracking,” Lecture for EE 6310: Optimal Control And Estimation, MST, Rolla.
- “Discrete optimal control,” Lecture for EE 6310: Optimal Control And Estimation, MST, Rolla.
- “Kalman filters,” Lecture for EE 6310: Optimal Control And Estimation, MST, Rolla.
- “Dynamic programming,” Lecture for EE 6310: Optimal Control And Estimation, MST, Rolla.
- Several presentations at conferences: ACC, CDC, IJCNN, and SSCI.

Funded Projects

1. **2024–27.** Learning-Enabled Collaborative Autonomy for Networked CPS, AFOSR, DOD, PI: Vignesh Narayanan, \$599,996.
2. **2024–27.** Vulnerability Characterization of Learning-Based Controllers in Networked Cyber-Physical Systems, NSF, PI: Vignesh Narayanan, \$300,027.
3. **2023–26.** Collaborative Autonomy in Uncertain Environments: Exploring Vistas Beyond Consensus, Naval Engineering Education Consortium, Naval Surface Warfare Center, Panama City, PI: Avimanyu Sahoo (UAH), Co-PI: Vignesh Narayanan, \$448,751 (my share \$218,914).
4. **2023–25.** EAGER: Knowledge-guided Neurosymbolic AI with Guardrails for Safe Virtual Health Assistants, NSF, PI: Amit Sheth, Co-PI: Vignesh Narayanan, \$200,000 (my share \$66,600).
5. **2024–25.** Next-Generation Multiagent-Based Simulation System in support of the ICS-INFEST Project, UMD, ARLIS, DOD, PI: Michael Huhns, Co-PIs: Biplav Srivastava and Vignesh Narayanan, \$75,000 (my share \$25,000).
6. **2022–23.** Information Competition Simulator (ICS), UMD, ARLIS, DOD, PI: Michael Huhns, Co-PI: Vignesh Narayanan, \$135,170 (my share \$106,070).
7. **2023–24.** Information Competition Simulator (ICS) – Phase II, UMD, ARLIS, DOD, PI: Michael Huhns, Co-PI: Vignesh Narayanan, \$59,957 (my share \$24,000).

Service Activities and Memberships

Organizational Service

- Organizer: Special session – “Systems-Theoretic Approaches to Learning 4.0: From classical to quantum neural networks,” World Congress on Computational Intelligence, 2026.
- Organizer: Minisymposium – “Systems-Theoretic Approaches to Learning: Applications to Estimation and Control 3.0,” SIAM Conference on Applications of Dynamical Systems (DS25), Denver, 2025.
- Organizer: Special session – “Systems-Theoretic Approaches to Learning: Applications to Estimation and Control 2.0,” International Joint Conference on Neural Networks, Rome, Italy, 2025.
- Organizer: Minisymposium – “Systems-Theoretic Approaches to Learning: Applications to Estimation and Control,” SIAM Conference on Mathematics of Data Science, Portland, 2024.
- Co-organizer: Special session at the World Congress on Computational Intelligence, 2022 – Data-Efficient Multi-Agent Reinforcement Learning for Control, Automation, and Population Systems Optimization.
- Organizer: AI Institute retreat and poster sessions, October 2022.

Committee Membership

- Member, Graduate Committee, Department of Computer Science and Engineering, University of South Carolina (since November 2022).
- Member, Graduate Admissions Sub-committee, Department of Computer Science and Engineering, University of South Carolina (since November 2022).

Professional Memberships

- IEEE Senior Member since 2023; IEEE Member 2017–2023; IEEE Student Member 2014–2017 (Computational Intelligence Society, Control Systems Society, and Systems, Man, and Cybernetics Society since 2015).

Editorial Roles

- Associate Editor, *IEEE Transactions on Neural Networks and Learning Systems* (since July 2023).
- Associate Editor, *IEEE Transactions on Cybernetics* (since January 2024).

Reviewing

- Reviewer: MIT Press; NSF proposal panelist; Veni-NWO, the Dutch Research Council; External doctoral dissertation examiner – Bharathiar University, India.
- Reviewer (Web of Science Researcher ID: I-9036-2019): *Nature Scientific Reports*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Cybernetics*, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, *IEEE Transactions on Automatic Control*, *IEEE Transactions on Control Systems Technology*, *IEEE Transactions on Industrial Electronics*, *IET Control Theory and Applications*, *Mechanical Systems and Signal Processing*, *Automatica*, *Neurocomputing*, *Journal of Franklin Institute*, *International Journal of Robust and Nonlinear Control*, *Nonlinear Analysis: Hybrid Systems*, *Optimal Control Applications and Methods*, and *Engineering Applications of Artificial Intelligence*; ICML 2025, NeurIPS 2025, AAAI 2026, ICLR 2026.

Doctoral Dissertation Committee Member

- Manas Gaur, USC, 2022 – Neurosymbolic AI and knowledge-infused learning
- Yao-Chi Yu, WashU, 2023 – Reinforcement learning and health informatics
- Ruwan Wickramarachchi, USC, 2023 – Neurosymbolic AI and scene understanding
- Yuan-Hung Kuan, WashU, 2024 – Learning and control
- Siva Likitha Valluru, USC, 2024 – Group recommendation systems
- Revathy Venkataramanan, USC, 2024 – Neurosymbolic AI and process knowledge infusion for recommendation systems
- Usha Lokala, USC, 2025 – Neurosymbolic AI and process knowledge infusion for social media data
- Vipula Rawte, USC, 2025 – Detecting and quantifying hallucination in large-language models
- Dingyang Chen, USC, 2025 – Multi-agent reinforcement learning
- Kaushik Roy, USC, 2025 – Neurosymbolic AI and process knowledge infusion
- Abir Hossen, USC, 2025 – Causal learning and multi-agent system design
- Hasibul Amin, USC, 2025 – Cross-layer design and optimization of analog in-memory computing systems
- Kausik Lakkaraju, USC, 2025 – AI rating
- Alireza Bagheri Rajeoni, USC, 2025 – Application of ML for vascular system analysis
- Jianhai Su, USC, 2026 – Sample-efficient reinforcement learning
- Chathurangi Shyalika, USC, 2026 – Neuro-symbolic AI for time-series data
- Andrew Smith, USC, 2026 – AI for time-series data with applications to activity recognition

Master's Thesis Committee Member

- Diksha Srishyla, USC, 2025 – Analyzing functional and effective connectivity metrics with EEG data
- Mark Shperkin, USC, 2025 – Pose estimation from swimming videos
- Kuba Jerzmanowski, USC, 2026 – ML for human-activity detection

Internships and Industrial Research Projects

- Research Intern, RCI, Defence Research and Development Organization, Hyderabad, India *Jul 2013 – May 2014*
- Research Assistant, National Institute of Technology, Kurukshetra, India *Sep 2013 – Dec 2013*
- Research Intern, RCI, Defence Research and Development Organization, Hyderabad, India *Dec 2011 – Apr 2012*

Students at USC

Graduate Students

1. Bharath Chandra Muppasani – Ph.D. Computer Science – Dynamic planning – Jointly advised by Prof. Biplav Srivastava (2022 –)
2. Erik Connerty – Ph.D. Computer Science – Quantum machine learning and networks (2023 –)
3. Protik Nag – Ph.D. Computer Science – Continual learning, graph learning, and explainability (2023 –)
4. Ritirupa Dey – Ph.D. Computer Science – Human-in-the-loop systems and collaborative autonomy (2024 –)

Undergraduate Students

1. Risha Patel – (Fall 2024) – Robotics, navigation and control
2. Sebastian Gaviria – (Spring 2025) – System support, robotics
3. Allan Paiz – (Spring 2026) – Robotics

Past Collaborations – Graduate Students

1. Sai Teja Paladi (M.S. Computer Science) – Jointly advised by Prof. Biplav Srivastava (2022–23)

Past Collaborations – Undergraduate Students

1. Gavin Hewitt – (Fall 2023) – Multi-agent modeling and support for ICS
2. Erik Connerty – (Fall 2022, Spring 2023) – Multi-agent modeling and support for ICS
3. Peyton Tucker – (Summer 2021) – Ontology and knowledge graph for ICS
4. Austin Cappuccio – (Fall 2022, Spring 2023) – Software development for ICS
5. Rae Jones – (Fall 2023 – Spring 2025) – Multi-agent modeling
6. Ayush Parambath – (Fall 2024 – Spring 2025) – Robotics, vision

Past Collaborations – Interns

1. Sanjay Chandrasekar, Rising Junior at Monta Vista High School, Cupertino, California (Fall 2022) – Natural language processing
2. Chandrika Singh Jadon, BITS Pilani, Goa, India – Machine learning and vision

Graduate Students Mentored Before Joining USC

1. Wei Miao (WashU) – Computational mathematics and ensemble control
2. Yao-Chi Yu (WashU) – Reinforcement learning and health informatics
3. Wei-Cheng Jiang (WashU) – Multi-agent reinforcement learning
4. Nathan Szanto (MST) – Event-triggered control for quadrotors
5. Yuan-Hung Kuan (WashU) – Learning and control
6. Liang Wang (WashU) – Data science and health informatics